

The real deal

I recommend you take a back-up of all your personal information on the memory card or a computer before you move forward with the actual rooting procedure. The process for rooting a phone or tablet varies greatly, depending on the device and the operating system of the computer used in the process.

There are several third-party applications, especially for the Windows, Mac and Linux operating systems, which allow you to connect the device and root without much intervention. Some of the popular applications for this process are Once Click Root, CF-Auto-Root and Super One Click. Most of these applications provide step-by-step instructions (sometimes with images) on how to root your device. Based on your device, the process will require you to press several buttons and reboot. As the steps vary from device to device, you need to search the Internet to find the best application—one that has had the most success with your device. Almost all these applications install the Superuser app that helps in granting super-user permissions or root access to other apps. There are also several ways to root a device using terminal commands from a Linux distribution. However, this procedure is only for experts. To verify if you have gained root access of your device, you can use apps such as Root Checker by joeykrim and Check Root by Avarna Software.

What happens next?

Once the rooting procedure is complete, you will see an app named Superuser, SuperSU or something similar among other installed apps on your device. Open the Superuser app and update it to obtain the latest version.

Superuser and SuperSU

As and when another app requests for super user permissions, the SuperSU app gives you a pop-up to ask whether or not you want to grant such access. Based on the option you choose, you can have SuperSU prompt you at a later time or not prompt you for access till the app is updated. Ensure that you provide root access only to trusted apps and revoke permissions for those you are unsure of. In addition, it is a good practice to have an app ask for permissions every time. This allows you to keep track of the apps yourself, in addition to the logs maintained by SuperSU. Superuser, SuperSU and other similar apps also allow you to unlock additional paid features and improve the functionality further. However, for a basic user, the free features would suffice.

Busybox

Called the Swiss army knife of embedded Linux operating systems, Busybox is actually a stripped down version of powerful Linux tools made available for the Android operating system. You can visit <http://www.busybox.net/> for more information on this project. If you're into tweaking the device even further, you will need to make use of the several tools provided in this bundle. Some custom ROMs provide

Busybox just as they do with SuperSU or Superuser. If not, there are several Busybox installers available on the Google Play app store, the most popular one being the BusyBox app by Stephen (Stericson). Most of the commands are the same as their counterparts on other operating systems.

SQLite

If you're into heavy duty tinkering, you will require specialised tools. SQLite is a relational database management system that is used by several apps running on Android. There are several apps available on the Google Play store and on F-Droid that allow you to install the SQLite3 binary application on your device. Once this is installed, you can use database viewers and editors as per your convenience. I would like to recommend the 'SQLite Installer for Root' app by Ptoft in case you plan to use a stable and secure installer.

The terminal

People with knowledge of the Linux command line will feel right at home in the terminal emulators available for Android. The Google Play app store showcases several terminal emulator apps that allow you to run shell level commands in Android. Some of the best terminal emulators are available on F-Droid, an open source repository of Android apps. Terminal emulators are also useful for running Busybox commands.

File explorers

Several file explorers, such as Root Explorer, allow users to view and edit Android system files if you grant them super-user permissions. You can also change file and folder permissions with the help of such applications. Most of these apps also allow you to connect to cloud storage and seamlessly work with archives and other such files.

Bye bye bloatware!

As discussed before, rooting allows you to remove the bloatware provided by your hardware manufacturer that you couldn't get rid of previously. Apps such as Root App Delete are available on the Google Play store and help you free up space. However, you are advised not to remove or disable system apps as they may hamper the functionality of your device. Similar to bloatware, unwanted ads is another problem that most people face. To block ads, these apps require root access to write into the hosts' folder in the Android system.

In addition to the above mentioned entities, there are many more avenues to be explored. Rooting, if done responsibly, opens new doors and adds tons of functionality to your devices. 

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The author is an open source enthusiast who loves to read up on science and technology. A science student herself, she has spent hours tinkering with several Linux distributions as well as Android, and wishes to share her knowledge in the true spirit of open source.